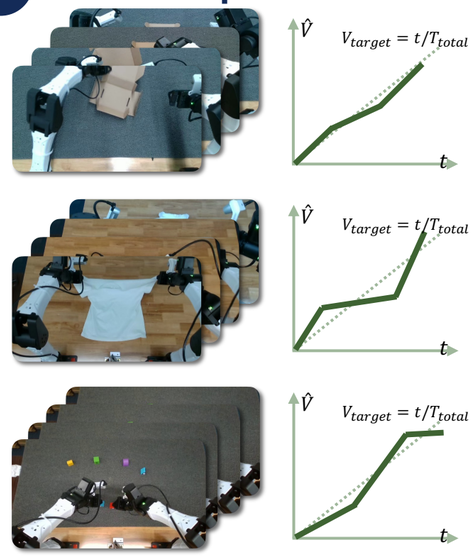
Training Pipeline

1 Offline Expert demo



Behavioural Cloning

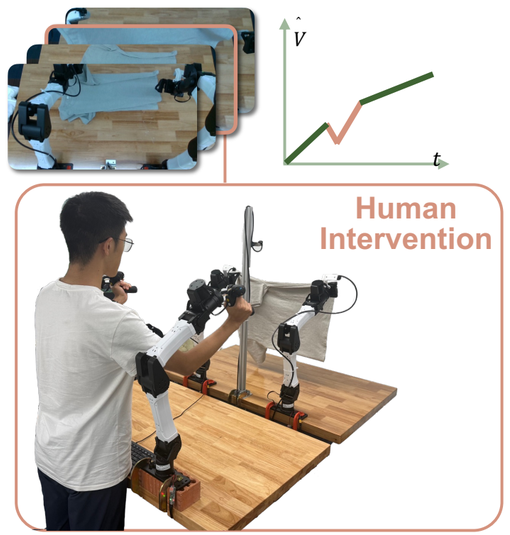
π Action Expert π_0

MC Progress Estimation

V Value Expert V_0

Deploy

2 Offline RL Warmup



Advantage Conditioning

π Action Expert π_1

TD Value Warmup

V Value Expert V_1

3 Online DAgger

4 Learn from Experience

Human Intervention

Autonomous Rollout

Deployment Runner Round k

π Action Expert π_k

Deploy

Intervention-timestep Truncation

Sparse Terminal Reward

Value Cache

V Value Expert V_k

Value Annotation

Advantage Labeling

Annotated Rollout

Online Trainer Round k

π Action Expert π_{k+1}

Adv. FM

1:1:1 Batch Sample

V Value Expert V_{k+1}

TD

1:1 Batch Downsample

Mixed Replay Buffer

History Intervention

Offline Demo

History Rollout

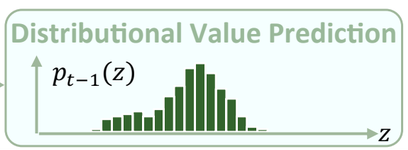
Current Rollout

Merge & Publish

Model Architecture

Distributional Value Head

Distributional Value Prediction



$\hat{V}_{t-1} = \mathbb{E}[p_{t-1}]$

Value Backbone

Readout Token

History Vision Encoder

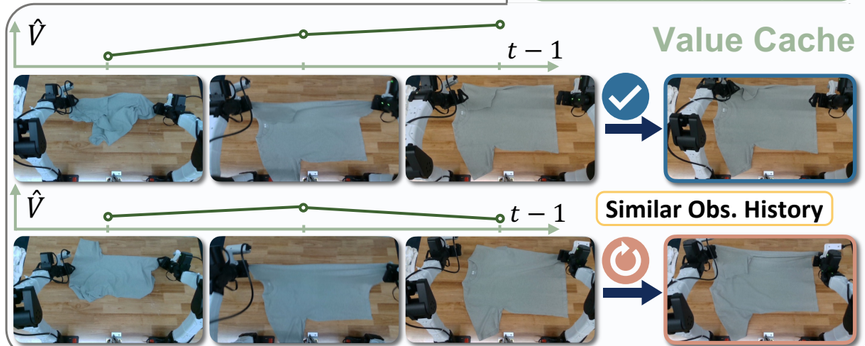
Task



$I_{t-H_V}, \ldots, I_{t-2}, I_{t-1}$

$\hat{V}_{t-H_\pi}$ ... $\hat{V}_{t-2}$ $\hat{V}_{t-1}$

Value Cache



VLM Backbone

History Vision Encoder

History Proprio Encoder

Cached Value Encoder

Task

Advantage token



S_{t-H_π} ... S_t

$\hat{V}_{t-H_\pi}$ ... $\hat{V}_{t-1}$

o_t

Action Chunk

Action Transformer

$\times N$

Noised action